Which Simple Shapes Wear Simple Sweaters?

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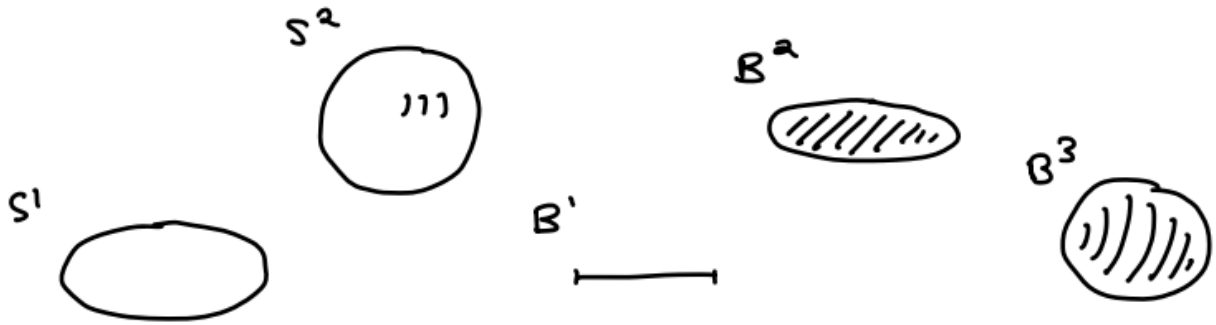
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Central Question

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1 Dimension

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It bounds the disk. This is simple because it's contractible and has no loops.

2 Dimensions

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Each such manifold encloses a solid, 3D version of itself. This is the "simplest" (oriented) manifold it can bound, in terms of number of loops. *So if a manifold bounds a "simple" manifold, it should also be "simple" itself.*

Central Question II

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Topological Invariants

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- fundamental group;
- homology groups.

Groups

Many useful topological invariants are *groups*.

Definition

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Definition

A group G is **abelian** if for all $a, b \in G$,

$$ab = ba$$

Fundamental Group

Definition

Consider a manifold X, and a point $p \in X$. The **fundamental group** $\pi_1(X, p)$ is a group whose:

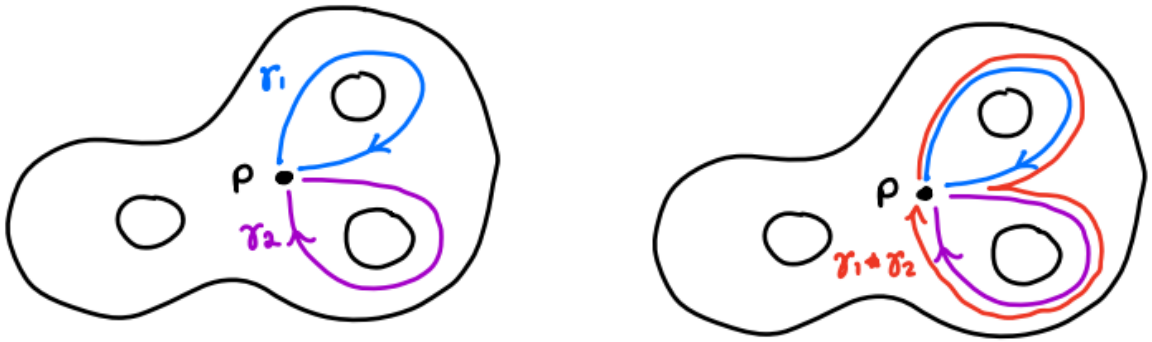
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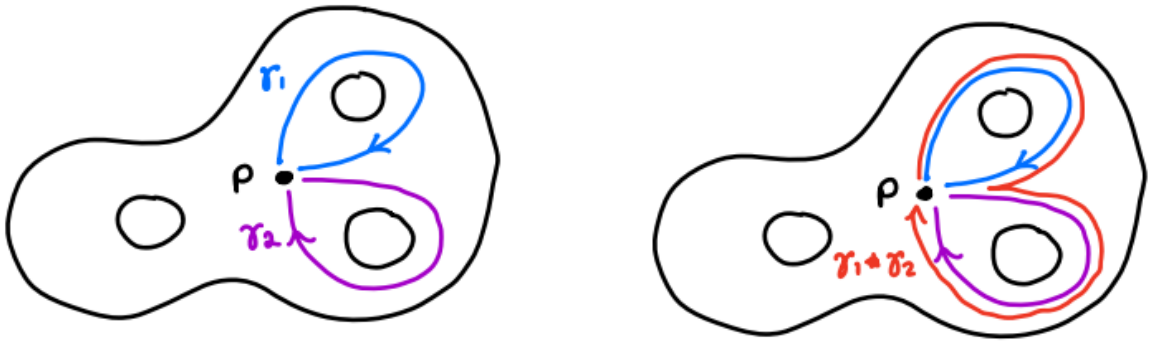


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Remark

If X is connected the choice of p doesn't matter, so we use $\pi_1(X)$.

Homology Groups

Definition

Consider a manifold X. The **first homology group** $H_1(X)$ is a group where:

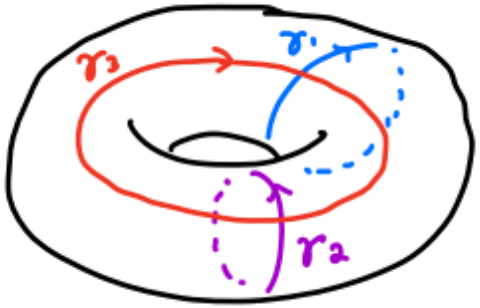
- the elements are "sums of loops", and the group law is adding sums of loops together;
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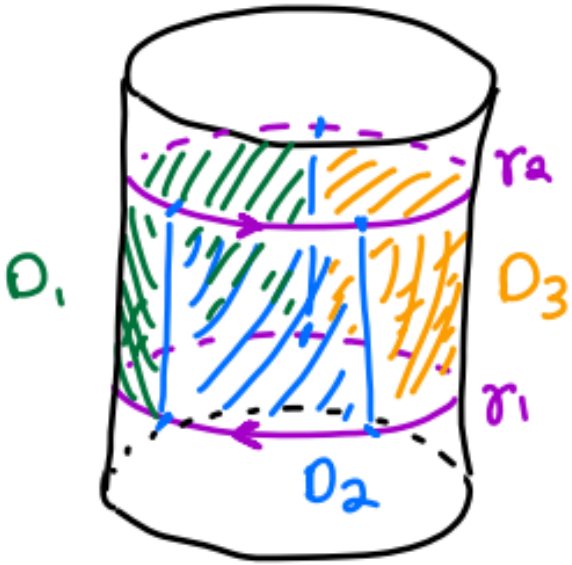
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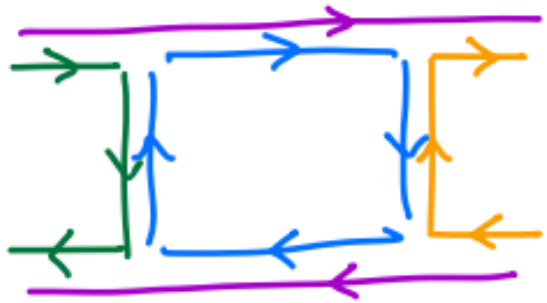
$$\alpha = 3[\gamma_1] - 2[\gamma_2] + 4[\gamma_3]$$

Homology Groups

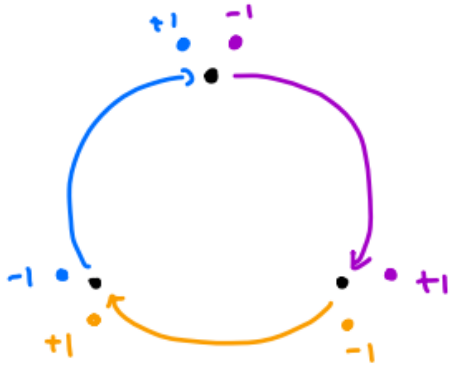


$$\gamma_1 + \gamma_2 = \partial(D_1 + D_2 + D_3)$$

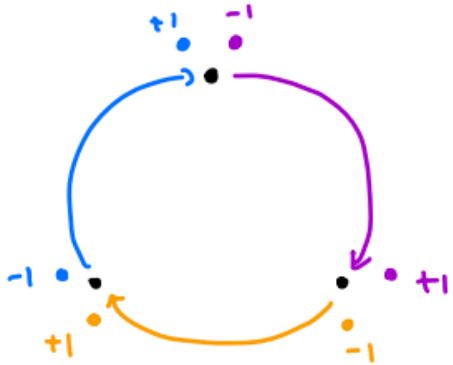
$$\Rightarrow \gamma_1 + \gamma_2 = 0$$



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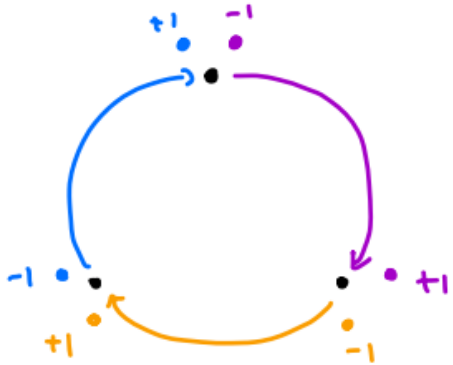
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Remark

Extend this definition to arbitrary dimension by replacing "loops" with sums of n-balls whose boundaries add to 0, and set such sums to be 0 if they are the boundaries of a sum of $n+1$-balls. This gives you $H_n(X, \mathbb{Z})$.

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Remark

$H_n(X)$ is abelian, hence of the form $\mathbb{Z}^m \oplus \bigoplus_i \mathbb{Z}/a_i$. The number of $\mathbb{Z}$s in H_n is called b_n, the nth ***Betti number****.*

Central Question III

When does a simple closed 3-manifold bound...

- a manifold with $\pi_1 = 0$?
- a manifold with $b_k = 0$?

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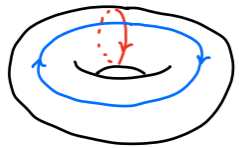
When does a simple closed 3-manifold bound a...

- simply connected manifold?
- a manifold with $b_k = 0$?

Torus Topology

To explain our choice of 3-manifold, we will first look at the topology of a torus.

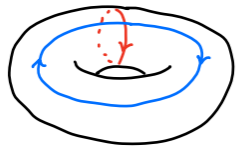
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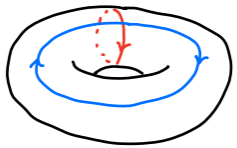


- A nonzero vector in $\mathbb{Z}^2$ which is not a positive multiple of another vector is called **primitive**.

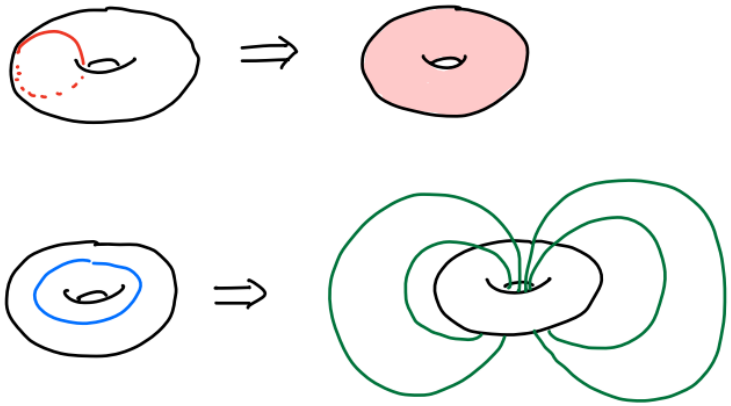
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- A nonzero vector in $\mathbb{Z}^2$ which is not a positive multiple of another vector is called **primitive**.
- To specify a **solid torus**, specify a primitive loop on the torus to be a meridian. All other primitive loops form longitudes.



Dehn Surgery

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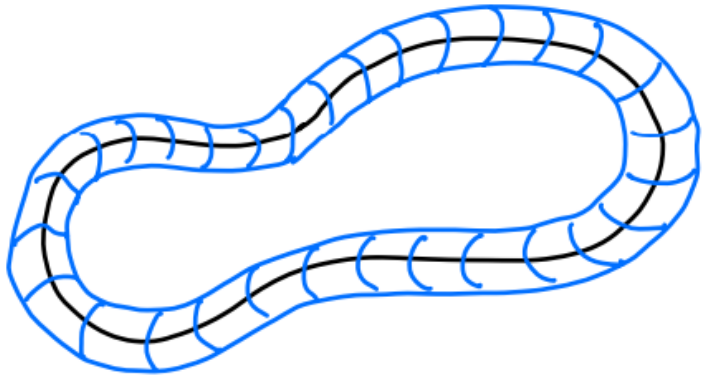
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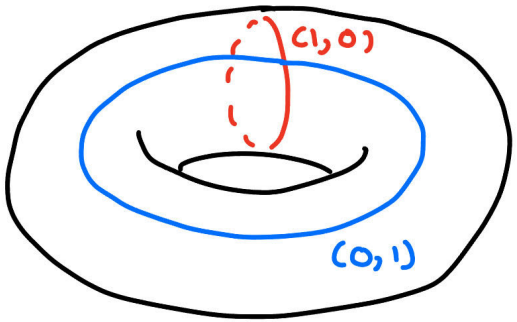
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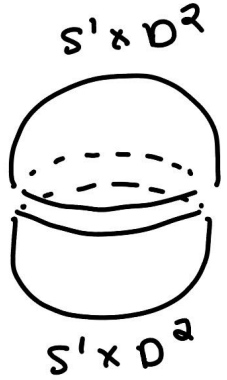
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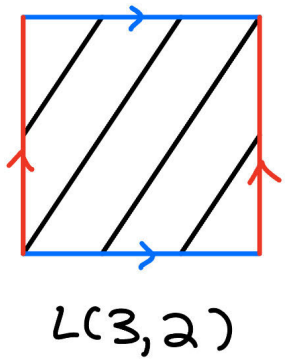
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Final Question

Which lens spaces $L(p, q)$ bound a 4-manifold with $b_2 = 0$?

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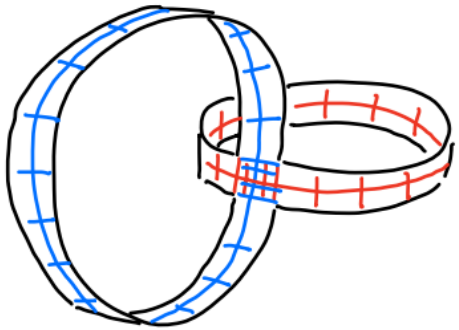
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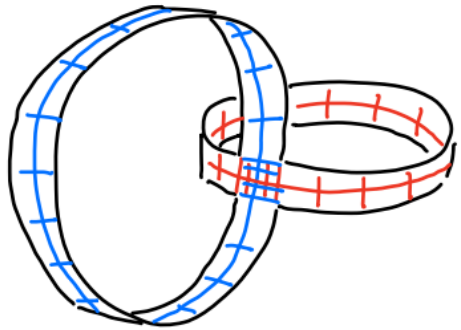
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For every lens space $L(p, q)$ there exists a canonical sphere plumbing $X(p, q)$ with boundary $L(p, q)$. The construction of $X(p, q)$ depends on the continued fraction expansion of p/q, so it can get arbitrarily complicated.

Intersection Form

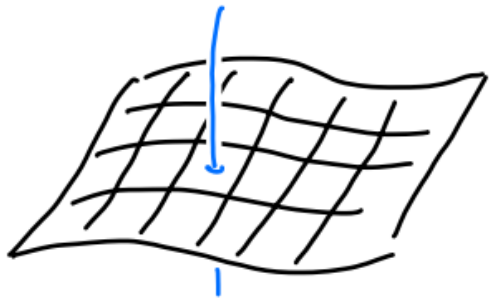
We will need one more invariant to answer our question.

- On a (closed oriented) 4-manifold X, each class in H_2 is represented by some embedded submanifold.
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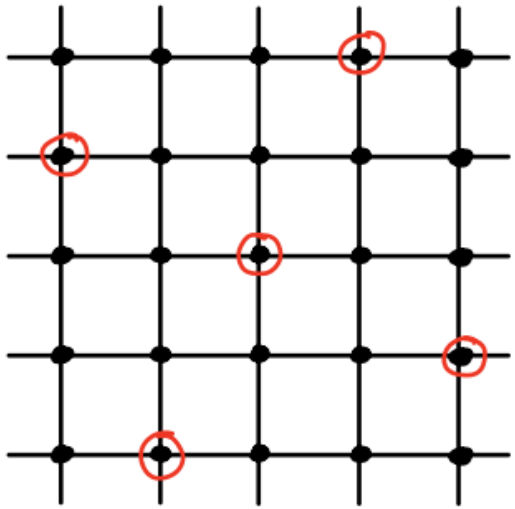
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- The lattice formed by $\mathbb{Z}^n$ with the usual inner product is denoted D^n.



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Both the manifolds above are closed and have definite intersection form. Thus by Donaldson's diagonalization theorem, both intersection lattices are of the form D^n for some n. Thus the intersection lattices for both $X(\pm p, q)$ must embed in some D^n of the same rank.

Answer

Definition

A reduced fraction p/q is good if $p = m^2$, and q is of one of the following forms:

1. $mk \pm 1$ for some $k \leq m$, $(k, m) = 1$;
2. $d(m \pm 1)$ for some positive divisor d of $2m \mp 1$;
3. $d(m \pm 1)$ for some positive odd divisor d of $m \pm 1$.

Theorem (Lisca, 2007)

Let $\mathcal{R}$ be the set of reduced fractions p/q such that there exists an r with p/r good, and $r \equiv \pm q^{\pm 1} \bmod p$. Then $L(p, q)$ bounds a 4-manifold with $b_2 = 0$ (in fact, a rational homology ball) iff $p/q \in \mathcal{R}$.

Next Directions

Which lens spaces $L(p, q)$ bound a 4-manifold with $b_2 = 1$?

Thanks!



References



Paolo Lisca
Lens spaces, rational balls and the ribbon conjecture.
Geom. Topol. 11 (1) 429 - 472, 2007.